

Arvind Verma

Associate Director Software Engineering | AI Transformation Leadership | Product Engineering

402-378-8550 | arvind_verma50@yahoo.com | [GitHub](#) | [LinkedIn](#) | [TechnologyBlog](#)

EXPERIENCE SUMMARY

Technology leader with 18 years of experience providing leadership, guidance, oversight, and mentorship in software engineering, product engineering, data engineering, multi-cloud environments (AWS, GCP, and Azure), AI, and Agentic AI application development, from ideation through production and enterprise-scale rollout. Adept at shaping long-term Engineering strategy and roadmap, cross-functional leadership, resource and capacity planning, scaling high-performing engineering teams, establishing governance frameworks, and partnering with senior executives to deliver measurable value. Deep knowledge of the financial domain, security analysis (both fundamental and technical), Data Packaging and Distribution, Monetization, and e-commerce applications targeting institutional customers.

SKILLS & EXPERTISE

AI & Machine Learning: Generative AI (GPT, Claude, Gemini), Agentic AI, Vertex AI, Google Agent Framework, RAG, MCP, Prompt Engineering, ML Model Development & Optimization, AI Governance & Responsible AI.

Software & Full-Stack Engineering: Architecture and Design Patterns, Microservices Architecture, API Lifecycle Management, Java Spring Boot, Python (Django/Flask), React/TypeScript, UI Frameworks, High-Availability & Distributed Systems, Reliability & Performance Engineering, Observability & Monitoring (Elastic, OpenSearch, Datadog), Test Automation (Selenium, Fitnesse, Cucumber).

Data Engineering & Platforms: Enterprise Data Engineering, Data Architecture & Modeling, Data Governance, Medallion Architecture, ETL/ELT, Big Data (Spark/Scala), Databricks, Snowflake, Delta Lake, Data Fabric, SQL/NoSQL (Oracle, Postgres, Aurora, SQL Server, MySQL, MongoDB, Redis).

Cloud Architecture & Infrastructure: AWS, GCP, Azure, Multi-Cloud Modernization, Serverless & Container Platforms, Cloud Security & IAM, Event-Driven Architecture, Infrastructure as Code (Terraform), CI/CD Pipelines (GitHub Actions, Azure DevOps, Jenkins), Unix/Bash, Cloud CLIs, DevSecOps & Security Tools (Dependabot, MEND, Fortify, WIZ, ArmorCode).

Leadership & Delivery: Engineering Leadership (teams 4–50), Org Scaling, Hiring & Mentoring, Technical Program Management, Cross-Functional Leadership, Agile (SCRUM, SAFe, Kanban), Enterprise Program Delivery.

WORK EXPERIENCE:

Associate Director, Software Engineering
Present

12/2019 –

S&P Global, Virginia, USA

AI, Agentic AI and Product Innovation

- Ideated, architected, and launched the Xpressfeed AI Accelerator, introducing agentic workflows that significantly accelerated migration of legacy data products.
- Drove enterprise adoption of GitHub Copilot, Claude Code, and other LLM-powered development tools through structured enablement and cost-optimization guidance.
- Delivered full-stack AI-enabled applications using React, TypeScript, Java Spring Boot, Python, SQL, and Scala.
- Led organization-wide AI transformation, including use-case identification, ROI evaluation, and post-implementation measurement.

Data Engineering, Analytics & Multi-Cloud Architecture

- Built and managed large-scale Databricks pipelines delivering data across Delta Sharing, Snowflake, S3, and SFTP channels.
- Directed migration of legacy ETL/ELT workloads into GCP, improving scalability, reliability, and cost efficiency.
- Architected a horizontally scalable data engineering ecosystem that significantly reduced packaging time and improved client onboarding.
- Designed and delivered microservice APIs in GCP, defining both application and infrastructure architecture.
- Provided architectural leadership across AWS and GCP for full-stack product development, AI workloads, and cloud-native data pipelines.

GCP Infrastructure, APIs & Data Sharing

- Designed and automated GCP infrastructure using Terraform, including storage, messaging, compute, IAM, and event-driven integrations.
- Led development of secure, enterprise-grade APIs exposing curated BigQuery datasets through governed data-sharing platforms.
- Defined end-to-end authentication and authorization patterns (OAuth, service account impersonation, token-based access, fine-grained IAM) to strengthen platform security.
- Established reusable Terraform modules and CI/CD patterns to standardize infrastructure provisioning across engineering teams.

Work Experience: Previous Employer

Tech Mahindra Limited – Engagement Manager, Omaha, Nebraska, USA 11/2011 to 12/2019

Key Contributions:

- **Directed cross-functional teams of 4–25** Software Engineers and SDETs in designing and delivering enterprise applications for Union Pacific Railroad’s Positive Train Control (PTC) initiative, a federally mandated safety program.
- **Scaled organisational capability**, hiring, onboarding, and mentoring 50+ engineers to support mission-critical systems, fostering a culture of accountability and technical excellence.
- **Delivered safety-critical applications** that ensured compliance with federal regulations, enhancing operational reliability and reducing risk across Union Pacific’s rail network.
- **Partnered with stakeholders** to gather requirements, prioritise backlog, and define product enhancements, aligning technical delivery with business objectives.
- **Established CI/CD best practices**, administering Jenkins pipelines to streamline deployments and strengthen test automation coverage.
- **Engineered automation frameworks**, developed Python scripts for lab testing and built Java/SpringBoot tools for JMS-based microservices, improving efficiency and reducing manual effort.
- **Contributed hands-on technical expertise** by writing unit-testable code for safety-critical applications, ensuring robustness and regulatory compliance.

Work Experience: Previous Employer

Impetus InfoTech India Pvt. Ltd – QA Test Engineer Indore, M.P., India 02/2010 – 11/2011

Key Contributions:

- Led a team of 4 QA Engineers, introducing automation practices that improved efficiency and reduced release cycles.
- Contributed to Android mobile application development and quality assurance, ensuring cross-platform reliability.
- Designed and executed comprehensive test strategies, strengthening product stability and user experience.

Work Experience: Previous Employer

Electronic Arts Games – QA Tester Hyderabad, Telangana, India 08/2008 – 02/2010

Key Contributions:

- Supported QA for bestselling titles, including FIFA, Wolverine, and The Sims, ensuring high performance and immersive user experiences.
- Designed and executed regression, UI, and functionality test cases, contributing to consistent product quality.

- Collaborated with game designers and developers, sparking an early interest in product design and engine development.

Education

- Master of Business Administration (Executive) in General Management from Symbiosis Institute of Business Management, Pune, Maharashtra, India. July 2012 - December 2014
- B.E in Computer Science and Engineering from Malwa Institute of Technology, Indore, M.P. October 2004 - June 2008.

Certification

- **AWS Cloud Practitioner: (expired; but actively using AWS services for the last 7 years)**
- **SAFe 5 Practitioner: Certified**